Year 10 · Chapter 4 · Statistics: answers (teacher copy)

4.1 Types of data (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: categorical
2. Ex 1b: numerical
3. Ex 1c: numerical
4. Ex 1d: categorical
5. Ex 2a: categorical, nominal (no order)
6. Ex 2b: categorical, ordinal (small → large has an order)
7. Ex 2c: numerical, discrete (counted)
8. Ex 2d: numerical, continuous (measured)

Easy

1. categorical
2. numerical
3. categorical
4. numerical
5. numerical
6. categorical
7. numerical
8. categorical
9. categorical
10. numerical
11. categorical
12. numerical

Medium

1. nominal
2. ordinal
3. nominal
4. ordinal
5. discrete
6. continuous
7. discrete
8. continuous
9. categorical, ordinal
10. numerical, continuous
11. categorical, nominal
12. numerical, discrete
13. e.g. “How many pets do you have?” (any question answered by counting)
14. categorical, ordinal: the answers are groups that have an order

4.2 Frequency tables (booklet p. 5)

WE DO examples (teacher)

1. Ex 1a: tallies of 7, 4, 2, 3
2. Ex 1b: apple 7, banana 4, grapes 2, mango 3
3. Ex 1c: apple
4. Ex 1d: 7 + 4 + 2 + 3 = 16 students
5. Ex 2a: tallies of 2, 7, 5, 1, 1
6. Ex 2b: 0–4: 2, 5–9: 7, 10–14: 5, 15–19: 1, 20–24: 1; total 16
7. Ex 2c: 0–4: 2 ÷ 16 × 100 = 12.5% · 5–9: 7 ÷ 16 × 100 = 43.75% · 10–14: 5 ÷ 16 × 100 = 31.25% · 15–19: 1 ÷ 16 × 100 = 6.25% · 20–24: 1 ÷ 16 × 100 = 6.25%
8. Ex 2d: 5–9 hours

Easy

1. 4
2. 9
3. 4
4. 3
5. 1
6. 20
7. 9
8. 4
9. Netball
10. Swimming
11. 25
12. 4

Medium

1. 1
2. 5
3. 8
4. 30–39
5. 4
6. 3
7. 12%
8. 24%
9. 32%
10. 20%
11. 12%
12. 100%
13. 9 ÷ 30 × 100 = 30%
14. 10–19: 3, 20–29: 5, 30–39: 2

4.3 Histograms (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 7 people
2. Ex 1b: 5–9 hours
3. Ex 1c: 2 + 7 + 5 + 1 + 1 = 16 people
4. Ex 1d: 5 + 1 + 1 = 7 people
5. Ex 2a: total 20 · 0–9: 6 ÷ 20 × 100 = 30% · 10–19: 8 ÷ 20 × 100 = 40% · 20–29: 4 ÷ 20 × 100 = 20% · 30–39: 2 ÷ 20 × 100 = 10%
6. Ex 2b: columns that touch, heights 6, 8, 4, 2; frequency up the side, minutes along the bottom

Easy

1. 8
2. 5
3. 20–29 cm
4. 40–49 cm
5. 30
6. 6
7. 4
8. 2
9. 30–39 points
10. 14
11. 6
12. 3

Medium

1. 25%
2. 30%
3. 20%
4. 24%
5. 60%
6. 24%
7. 35%
8. 25%
9. columns 1, 3, 6, 7, 3; most common 30–39
10. columns 5, 9, 4, 2; most common 3–5 hours
11. 25%
12. 20 values, 40%

4.4 Mean, median, mode and range

(booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 30 ÷ 5 = 6
2. Ex 1b: 3, 5, 7, 7, 8 → 7
3. Ex 1c: 7
4. Ex 1d: 8 − 3 = 5
5. Ex 2a: mean 30 ÷ 6 = 5, range 10 − 2 = 8
6. Ex 2b: 12, 15, 18, 22, 26, 29 → (18 + 22) ÷ 2 = 20
7. Ex 2c: mean 562 ÷ 13 = 43.2, median 38
8. Ex 2d: the two oldest ages (87 and 76) are outliers that pull the mean up

Easy

1. 12 ÷ 3 = 4
2. 20 ÷ 4 = 5
3. 150 ÷ 5 = 30
4. 25 ÷ 5 = 5
5. 1, 3, 5, 7, 9 → 5
6. 4, 8, 12 → 8
7. 2, 4, 6, 7, 9 → 6
8. 11, 13, 15, 18, 20 → 15
9. mode 4, range 8 − 2 = 6
10. mode 9, range 9 − 1 = 8
11. mode 7, range 10 − 2 = 8
12. mode 12, range 18 − 11 = 7

Medium

1. 3, 5, 8, 10 → (5 + 8) ÷ 2 = 6.5
2. 4, 7, 9, 12 → (7 + 9) ÷ 2 = 8
3. 11, 14, 16, 18, 20, 25 → (16 + 18) ÷ 2 = 17
4. 1, 2, 5, 6, 8, 9 → (5 + 6) ÷ 2 = 5.5
5. 14 ÷ 3 = 4.7
6. 51 ÷ 4 = 12.8
7. 31 ÷ 6 = 5.2
8. 137 ÷ 6 = 22.8
9. mean 75 ÷ 5 = 15, median 7; the median (the outlier 49 pulls the mean up)
10. mean 260 ÷ 6 = 43.3, median 35; the median (the outlier 90 pulls the mean up)
11. mean 35 ÷ 7 = 5, median 5, range 8 − 2 = 6
12. total = 4 × 15 = 60; 60 − 44 = 16

4.5 Stem-and-leaf plots (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: 31, 31, 37, 38
2. Ex 1b: 24
3. Ex 1c: 31
4. Ex 1d: 11
5. Ex 2a: 1 | 5 9; 2 | 0 4 8 9; 3 | 2 4 7 8 8; 4 | 2 9; 5 | 0
6. Ex 2b: (32 + 34) ÷ 2 = 33
7. Ex 2c: 50 − 15 = 35
8. Ex 2d: in the 20s and 30s (9 of the 14 scores)

Easy

1. 15
2. 42
3. 23
4. 11
5. 5
6. 4
7. 56
8. 95
9. 73
10. 14
11. 5
12. 95 − 56 = 39

Medium

1. 1 | 8 9; 2 | 3 6 9; 3 | 1 5 7 8; 4 | 1 4 5; median (31 + 35) ÷ 2 = 33; range 45 − 18 = 27
2. 1 | 2 4 8; 2 | 0 2 5 6 7 9; 3 | 1 3 5; median (25 + 26) ÷ 2 = 25.5; range 35 − 12 = 23
3. 11
4. the 6th value = 24
5. 40 − 12 = 28
6. 4
7. 15, 21, 24, 32, 36, 39, 50, 51, 57, 65, 73, 73, 82, 86; median (50 + 51) ÷ 2 = 50.5; range 86 − 15 = 71
8. there are 4 values (even), so the median is halfway between the two middle values: (15 + 18) ÷ 2 = 16.5

4.6 Symmetrical or skewed (booklet p. 13)

WE DO examples (teacher)

1. Ex 1a: symmetrical: both sides match
2. Ex 1b: positively skewed: the tail is on the right
3. Ex 1c: negatively skewed: the tail is on the left
4. Ex 1d: positively skewed: the tail is on the right
5. Ex 2a: positively skewed (the tail is on the right)
6. Ex 2b: 5–9 and 10–14 hours (12 of the 16 people)
7. Ex 2c: the mean (9.6 > 8.5)
8. Ex 2d: the few big values in the tail pull the mean up

Easy

1. symmetrical
2. positively skewed
3. negatively skewed
4. positively skewed
5. symmetrical
6. negatively skewed
7. positively skewed
8. symmetrical
9. positively skewed
10. negatively skewed
11. symmetrical
12. positively skewed

Medium

1. 12
2. 20
3. 15
4. 1
5. symmetrical; most in 20–29
6. positively skewed; most in 10–14
7. negatively skewed; most in 90–99
8. positively skewed; most in 2–3
9. the mean: the few very expensive houses (the tail) pull the mean up
10. outlier 12; mean with it 43.7, without it 50

4.7 Time-series graphs (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 19 °C
2. Ex 1b: 25 °C at 2 pm
3. Ex 1c: 12 °C at 6 am
4. Ex 1d: 4 pm
5. Ex 2a: upward
6. Ex 2b: downward
7. Ex 2c: no trend (it goes up and down)
8. Ex 2d: a downward linear trend: down 4 each time, so next 30

Easy

1. 22 ($22 000)
2. 5 ($5 000)
3. January
4. July
5. 30 − 6 = 24 ($24 000)
6. November
7. 25 cm
8. 75 cm
9. week 4
10. 60 − 40 = 20 cm
11. 82 − 5 = 77 cm
12. week 3 to week 4 (20 cm)

Medium

1. upward
2. downward
3. no trend
4. upward
5. upward
6. 2021
7. 730 − 620 = 110
8. 2021 to 2022 (+30)
9. about 745 (it rises about 15 a year lately)
10. 680
11. a downward linear trend (down $3 000 a year); year 5: $8 000
12. (180 − 120) ÷ 6 = 10 items